

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. On a perturbative Maxwellian phase chart we prove a nonautonomous kinetic energy estimate and construct the joint density–contact Gaussian field directly from microscopic marked cumulants. The collision noise acts on the exact dual coefficient $\Delta p + \psi$, including direct contact variance. The covariance kernel, Cameron–Martin domain, second epi-derivative and stopping-time process tightness are then identified without omitting the contact coordinate.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: THE JOINT COLLISION GAUSSIAN FIELD AND ITS CLOSED INVERSE FORM

The primitive noise lives on collision space. Density and contact fluctuations are two images of the same Gaussian random measure, and a dual pair (p, ψ) sees the single coefficient $\Delta p + \psi$. Thus pure contact tests retain their Poisson diagonal variance.

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1.1. The perturbative phase chart and energy space. Fix a smooth Maxwellian M and a compact chart of exposed paths satisfying

$$cM \leq f(t, x, v) \leq CM, \quad \|f - M\|_{W^{2,\infty}(M^{-1})} + \|q - 1\|_{W_w^{1,\infty}} \leq \delta, \quad 0 < q_- \leq q \leq q_+.$$

The chart is obtained by restricting the B2 real-source ball so that its first and second pressure derivatives remain in these bounds.

Let $\mathbf{V} \subset \mathbf{H} \subset \mathbf{V}^*$ be the kinetic Gelfand triple whose norm contains transport derivatives, the weighted collision Dirichlet form, the five spatially dependent macroscopic coefficient fields, and the B2 normal trace graph.

Theorem 1.1 (Nonautonomous perturbative kinetic energy estimate). *For δ and T sufficiently small, the linearized biased equation*

$$\partial_t u + v \cdot \nabla_x u = \mathcal{L}_{f,q} u + F, \quad u(0) = u_0,$$

has a unique solution in $L^2(0, T; \mathbf{V}) \cap C([0, T]; \mathbf{H})$ and

$$\sup_{t \leq T} \|u_t\|_{\mathbf{H}}^2 + c \int_0^T \|u_t\|_{\mathbf{V}}^2 dt \leq C \left(\|u_0\|_{\mathbf{H}}^2 + \int_0^T \|F_t\|_{\mathbf{V}^*}^2 dt \right).$$

The trace pairing is continuous on this solution space.

Proof. For $q = 1$ and $f = M$, the symmetric linearized collision form is coercive modulo the five velocity invariants. Projecting onto those invariants gives spatially dependent mass, momentum, and energy fields. Add the standard transport–moment cross terms to the energy; integration by parts and the moment equations control the macroscopic component without pretending it is five scalar constants. The $W^{2,\infty}$ coefficient perturbations and $q - 1$ terms are form-bounded with relative bound $O(\delta + T)$. Absorb them and use Lions’ theorem for nonautonomous forms. The B2 Green graph estimate controls the normal trace by the graph part of $\mathbf{V}$. $\square$

1.2. The joint Gaussian random measure. Let $\mathfrak{C}_T$ be collision space and define the intensity

$$d\mathfrak{a} = q(t, x, v, v_*, \omega) dA_f.$$

Let W be the centered isonormal Gaussian random measure on $L^2(\mathfrak{C}_T, \mathfrak{a})$, independent of the prepared initial Gaussian field G_0 . For a density test p and a direct contact test ψ , write

$$W(p, \psi) = \int_{\mathfrak{C}_T} (\Delta p + \psi) dW.$$

Theorem 1.2 (Prepared joint density–contact Gaussian process). *There is a unique centered Gaussian pair (u, Ξ) satisfying*

$$d\langle u_t, p_t \rangle = \langle u_t, (\partial_t + v \cdot \nabla_x + \mathcal{L}_{f,q}^*) p_t \rangle dt + \int \Delta p_t dW_t,$$

with initial law G_0 , and

$$\Xi(dz) = W(dz) + D(qA_f)[u](dz).$$

For two dual pairs (p, ψ) and (p', ψ') ,

$$d\langle M(p, \psi), M(p', \psi') \rangle_t = \int q(\Delta p + \psi)(\Delta p' + \psi') dA_f dt.$$

In particular a pure contact test has variance $\int q\psi^2 dA_f > 0$ unless it vanishes in $L^2(\mathfrak{a})$.

Proof. The stochastic convolution with the evolution family from Theorem 1.1 defines u . Linearizing the collision intensity qA_f gives the drift response in Ξ , while the same random measure W is the diagonal fluctuation of actual contacts. Applying a dual pair combines the density jump and direct contact mark into $\Delta p + \psi$. The isometry of W gives the bracket and uniqueness. $\square$

1.3. Microscopic cumulants and stopping-time tightness. Let

$$u^\varepsilon = \sqrt{\mu_\varepsilon}(\pi^\varepsilon - f), \quad \Xi^\varepsilon = \sqrt{\mu_\varepsilon}(\Gamma^\varepsilon - qA_f).$$

Lemma 1.3 (Conditional localized cumulants). *For every finite family of density/contact tests and every bounded stopping time τ , uniformly for $0 \leq h \leq h_0$,*

$$\mathbb{E} \left[|Z_{\tau+h}^\varepsilon - Z_\tau^\varepsilon|^4 \mid \mathcal{F}_\tau \right] \leq C \left(h^2 + \frac{h}{\mu_\varepsilon} \right)$$

after stopping on one compact moment set; the stopped complement has exponentially small probability. Joint cumulants of order $k \geq 3$ are $O(\mu_\varepsilon^{1-k/2})$.

Proof. Condition on the microscopic state at τ and restart the deterministic flow. The B2 fixed-horizon trajectory expansion is uniform on compact moment/trace sets and localizes every connected diagram to the interval $[\tau, \tau+h]$. Pairings of two contact insertions give h^2 , while one coincident contact diagonal gives the necessary h/μ_ε term. Higher connected diagrams carry the usual power $\mu^{1-k/2}$. B2 exponential compact containment removes the stopped complement. This conditional estimate, rather than a deterministic-interval bound alone, is the Aldous input. $\square$

Theorem 1.4 (Microscopic joint process convergence). *The pair $(u^\varepsilon, \Xi^\varepsilon)$ converges in the weighted trace-distribution Skorokhod space to the Gaussian pair of Theorem 1.2. All density–density, density–contact, and contact–contact covariances are the second derivatives of the limiting B2/B1 pressure.*

Proof. Normal convergence of the marked pressure gives convergence of finite cumulants. Lemma 1.3 and Aldous–Mitoma give process tightness, including the vanishing jump size. Passing the exact microscopic balance to the limit identifies the density equation, and the marked contact coordinates identify the random measure W . The initial Schur-complement law comes from B1. Uniqueness in Theorem 1.2 identifies every subsequence. $\square$

1.4. The exact dual kernel and gauge. Let $\mathcal{Y}$ be the weighted dual pair space with endpoint traces. Define

$$\mathfrak{D}(p, \psi) = \Delta p + \psi, \quad \mathfrak{G}r = (r, -\Delta r),$$

and quotient additionally by endpoint/transport coboundaries whose complete weak-balance pairing vanishes. Let C denote the initial constraint map.

Theorem 1.5 (Exact covariance kernel). *A dual class has zero covariance if and only if*

- (i) *its initial source lies in the span projected out by the B1 constraint Schur complement;*
- (ii) *$\Delta p + \psi = 0$ in $L^2(\mathfrak{a})$; and*
- (iii) *its remaining endpoint/transport pairing is a closed weak-balance coboundary.*

Thus the identifiable contact cotangent is the class of $\Delta p + \psi$, and pure contact directions are not quotiented out.

Proof. The covariance is the sum of two nonnegative terms: the prepared initial covariance and the $L^2(\mathfrak{a})$ norm of $\Delta p + \psi$ transported through the linear evolution. Their vanishing gives (i)–(ii). The closed range of the weak balance operator on the Gelfand triple from Theorem 1.1 identifies the residual annihilator with (iii). Conversely each listed direction pairs to zero with the initial field and with W , so it has zero covariance. $\square$

1.5. The Cameron–Martin form and action tangent. Let Σ be the covariance operator of the quotient Gaussian process. Define the closed extended form

$$\mathfrak{q}(z) = \begin{cases} \|\Sigma^{-1/2}z\|^2, & z \in \text{Ran } \Sigma^{1/2}, \\ +\infty, & \text{otherwise.} \end{cases}$$

Theorem 1.6 (Second epi-derivative and Mosco limit). *The second epi-derivative of the B2/B1 action at a regular exposed path is $\frac{1}{2}\mathfrak{q}$. Finite-dimensional action Hessians Mosco-converge to this closed form on the quotient Gelfand space.*

Proof. For every finite projection, analytic pressure duality identifies the action Hessian with the inverse of the projected covariance. Realize all projections as conditional expectations of the single Gaussian pair in Theorem 1.2; the corresponding Cameron–Martin forms are increasing closed forms. Monotone convergence of closed quadratic forms gives the Mosco liminf and recovery on $\text{Ran } \Sigma^{1/2}$. Density of finite projections in that range and $+\infty$ outside it complete the proof. The direct second variation of the entropy action agrees on a core and hence has the same closure. $\square$

Theorem 1.7 (Round-thirteen B3 closure). *The prepared hard-sphere tangent is a joint density–actual-contact Gaussian process driven by one collision-space random measure. Its exact dual noise coefficient is $\Delta p + \psi$, its kernel*

is the closed balance/endpoint gauge above, and its action tangent is the Cameron–Martin form on $\text{Ran } \Sigma^{1/2}$.

Proof. Combine Theorems [1.2](#), [1.4](#), [1.5](#), and [1.6](#). □

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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